HBEPA Reference Architecture: Gate 2 Technical Implementation Manual

This module defines the technical implementation standards for **Gate 2 (Network Security & Session Persistence)** within the Home-Based Employment Preservation Accommodation (HBEPA) reference architecture.

1. Architectural Objective

The purpose of Gate 2 is to establish secure identity verification and remote electronic presence natively, satisfying corporate audit guidelines without employing continuous desktop tracking, physical geographical location surveillance, or invasive webcam monitoring.

2. Authentication Protocol Sequence

To log an active persistence block, an approved user endpoint must complete the following secure handshake sequence:

1. **1. Encrypted Tunnel Initiation:** The remote client establishes a secure IPsec or WireGuard-based Virtual Private Network (VPN) connection to the dedicated enterprise security gateway.
2. **2. Federated Identity Handshake:** The session routes through an Identity Provider (IdP) utilizing Security Assertion Markup Language (SAML 2.0) or OpenID Connect (OIDC).
3. **3. Hardware Multi-Factor Authentication (MFA):** Identity verification requires an explicit hardware security token (FIDO2 / WebAuthn standard compliant, such as a YubiKey or secure device-bound biometric module) to confirm user presence.

3. Session Persistence & Logging Standards

- ***Tamper-Evident Ledger:*** Once authenticated, the server infrastructure initializes a persistent network state log linked directly to the anonymized employee ID.
- ***Network Keep-Alives:*** The system monitors standard encrypted session keep-alive pings to track connection durability automatically.
- ***Data Segregation Boundaries:*** Server tracking metrics generate an unalterable audit log specifying the timestamp entry and exit values. These files are restricted entirely to security teams and map directly to the Gate 3 payroll logging interface.